

Sai Yasaswini Majety

Senior Backend & Cloud Infrastructure Engineer

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U.S. Work Authorization: H-4 EAD (No Visa Sponsorship Needed)

Professional Summary

Senior Backend and Cloud Infrastructure Engineer with **6+ years** designing and deploying high-availability distributed systems, in-flight fleet telemetry microservices, and automated multi-environment cloud platforms. Ex-Senior Software Engineer at **Viasat (5.7+ years)**, sustaining **99.99% network availability** for commercial airline fleets (including Delta Air Lines), cutting deployment cycle times by **65%** via CloudFormation and Jenkins automation, and executing live zero-downtime AWS RDS MySQL upgrades. Co-inventor of published **Indian Patent 202541026299** in parameter-efficient modular deep learning adapter architectures, driving an **80% compute reduction** in incremental fine-tuning. Production depth spanning Python (FastAPI, Django), Go, AWS (ECS, RDS, S3, CloudFormation, IAM), Docker, Kubernetes, Terraform, HashiCorp Vault, and OAuth 2.0 / Okta security federation.

Technical Skills

Core Languages: Python (FastAPI, Django, Flask, Pytest, Celery), **Go (Golang)**, SQL (PostgreSQL, MySQL), Bash/Shell, Groovy

Cloud Infrastructure: **AWS** (ECS, RDS Multi-AZ, S3, IAM, CloudFormation, CloudWatch, Lambda, VPC, Route 53), Terraform, Docker, Kubernetes, Helm

Distributed Systems & Data: Microservices architecture, high-throughput REST APIs, message queues (RabbitMQ, SQS, Celery), in-flight telemetry ingestion, event streaming, distributed caching (Redis)

Security & Identity Governance: Automated TLS/SSL certificate lifecycle management, **OAuth 2.0**, **Okta** identity federation, HashiCorp Vault, RBAC, zero-trust network boundaries, Palo Alto firewalls, Aruba controllers

Databases & Storage: AWS RDS MySQL (5.6, 5.7, 8.0), PostgreSQL, DynamoDB, MongoDB, schema migrations, read-replica scaling, connection pooling, Sphinx search, Apache Solr

CI/CD & Observability: Jenkins scripted pipelines, GitOps, GitHub Actions, Datadog, Prometheus, Grafana, CloudWatch Logs, canary releases, blue/green deployments

Applied AI Systems: Modular deep learning adapter routing (Patent 202541026299), RAG microservices, LiteLLM proxy, semantic vector caching

Professional Experience

Independent Applied Research

AI Systems & Cloud Infrastructure Researcher

Mar 2024 - Present

Houston, TX (Remote)

- Published **Indian Patent 202541026299** with the Indian Patent Office (IPO) for Modular Deep Learning Architecture, achieving an **80% compute reduction** (from 48 GPU-hours to 9.6 GPU-hours per domain) by designing dynamic adapter routing that eliminates catastrophic forgetting across continuous domain transfers.
- Bounded P95 semantic query latency to **sub-20ms** (from 820ms un-cached LLM round-trips) across 5,000+ test queries by building an asynchronous semantic vector caching layer in **FastAPI** with Redis and LiteLLM proxy gateway routing.
- Automated multi-tier cloud infrastructure provisioning across AWS ECS and EKS by authoring modular **Terraform** blueprints with strict IAM least-privilege boundary policies and zero-drift CI/CD validation.
- Containerized microservice workloads into production-ready multi-stage **Docker** artifacts, reducing container image footprint by **58%** (from 850MB to 357MB) and hardening runtime containers against non-root privilege escalations.

Viasat Inc.

Senior Software Engineer - DevOps & System Integration

Jun 2018 - Feb 2024

Chennai, India

- Sustained **99.99% network availability** across 1,200+ commercial airline aircraft (including Delta Air Lines fleets) by engineering resilient backend microservices in **Python (FastAPI)** and **Go** to handle real-time passenger access control and telemetry synchronization.
- Accelerated deployment frequency by **65%** (reducing release cycle times from 45 minutes to 16 minutes) by authoring modular **AWS CloudFormation** infrastructure templates and orchestrating automated multi-stage **Jenkins** pipelines in Groovy.
- Eliminated **100% of production certificate outage risks** across 80+ in-flight network gateways by engineering an automated TLS/SSL certificate lifecycle daemon that continuously monitors expiry, requests ACME renewals, and rotates keys without service disruption.
- Migrated enterprise authentication to **OAuth 2.0** and Okta identity federation across all internal administrative services, securing 10,000+ daily employee and operator authentications and earning formal Architecture Commendation.
- Executed live zero-downtime database upgrades of **AWS RDS MySQL** clusters (from 5.6 to 5.7 and 8.0) across Multi-AZ environments by designing rolling schema migration procedures, preserving active flight data continuity with 0 data loss.

- Won **1st Place** in the Viasat Global IoT Hackathon among 40+ engineering teams by prototyping an end-to-end edge telemetry ingestion gateway with automated anomaly detection for aircraft passenger-cabin IoT sensors.
- Reduced production incident mean-time-to-resolution (MTTR) by **42%** (from 38 minutes to 22 minutes) by instituting end-to-end logging, CloudWatch alerting, and automated diagnostic health checks across production gateways.

- Accelerated candidate resume search to **sub-second P95 latency** across 10,000+ technical resumes (from 18-second manual lookups to 420ms) by architecting CandEx 2.0 using **Python (Django)**, Sphinx search engine, and Apache Solr indexing.
- Eliminated 15+ hours of weekly manual recruiter screening by building bulk resume ingestion workflows and automated candidate status dispatchers, receiving formal Leadership Appreciation for enterprise platform impact.

Patents & Research Publications

- **Indian Patent Published (App. No. 202541026299):** *Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning*. Published in The Patent Office Journal No. 13/2025 by the Government of India / Indian Patent Office (IPO). Co-inventors: Sai Likhith Kanuparthi, Sai Yasaswini Majety. Classified under IPC: G06N 3/08, G06N 3/045, G06N 3/096, G06N 20/00.

Honors & Technical Recognitions

- **1st Place Winner - Viasat IoT Global Hackathon (2019):** Awarded 1st place among 40+ engineering teams for architecting in-flight IoT telemetry ingestion and edge analytics pipelines.
- **Viasat Spotlight Award - Automated TLS/SSL Security (2023):** Recognized by Core Infrastructure Group for eliminating certificate expiration risks across production aviation gateways.
- **Architecture Commendation - Enterprise Identity Gateway (2022):** Commended by Viasat Architecture Committee for zero-downtime OAuth 2.0 migration and centralized access governance.
- **Engineering Excellence - Zero-Downtime Cloud Migration (2022):** Awarded by Platform Operations for seamless Multi-AZ database upgrades with zero flight passenger disruption.
- **Dean's Merit Scholarship (Top 3% Distinction):** Conferred by SASTRA Deemed University for outstanding academic ranking across the engineering cohort (2016-2017).

Education

Shanmugha Arts, Science, Technology & Research Academy (SASTRA Deemed University) **Thanjavur, India**
Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering (EEE) - CGPA: 8.2 / 10.0 **Aug 2014 - May 2018**

- Honors: Dean's Merit Scholarship Recipient (Ranked in Top 3% of Engineering Cohort, 2016-2017); Graduated First Class with Distinction.
- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Control Systems, Digital Signal Processing.